Reactive Time-Domain Motion Planning via a Configuration-Independent Double-Integrator Backbone

Yongyan Cao

***Abstract*—Traditional motion-planning pipelines separate geometric path generation from time parameterization, making updates costly when obstacles or task objectives change. This paper presents a local predictive motion planner formulated directly in time. Planning on a virtual double-integrator backbone rather than a plant linearization keeps the state-transition matrix A_d constant across configurations. Prediction matrices are therefore precomputed offline, and online planning becomes a convex quadratic program (QP) with fixed dynamics structure.**

The planner jointly enforces velocity and acceleration limits and updates the trajectory without rebuilding a geometric path. This exchanges static-path time optimality for reactive constrained generation: time-optimal parameterization is 3–3.5× faster with position tracking, while an analytic velocity-law reference narrows the ratio to 1.06–1.32×. On a fixed grid, the planner generates $(q_d, \dot{q}_d)$ with hard-bounded, piecewise-constant $\ddot{q}_d$. A triple-integrator extension bounds jerk when continuous acceleration is required; in autonomous-vehicle lateral planning, it produces smooth steering within the tested actuator-rate envelope. FR3 benchmarks show fixed-size reactive updates and exact satisfaction of the declared kinematic bounds. On an Intel Xeon w7-2475X, the single-threaded Python harness records coupled-QP solve p95 of 3.87–4.49 ms and full-cycle p95 of 6.60–7.70 ms; full-cycle maxima of 12.26–18.48 ms identify the tail-latency target for hard 100 Hz deployment.

***Index Terms*—Motion planning, model predictive control, reactive replanning, real-time trajectory generation, double integrator, obstacle avoidance, time parameterization.**

I. Introduction

Modern robotic systems deployed in unstructured or human-shared environments must track planned trajectories with high precision while remaining responsive to dynamic workspace changes. Classical motion planning pipelines often address these demands sequentially. A geometric planner, such as OMPL [1], first searches for a collision-free path in configuration space. A separate time-parameterization stage, such as Time-Optimal Trajectory Generation (TOTG) [2], then maps this spatial path into a timed trajectory through an arc-length-to-time transformation $s \to t$. This separation is effective for static scenes, but it creates a structural latency issue in reactive settings: when obstacles or task objectives change, the system must update the geometric path, the time law, or both before a new command can be executed. Even when incremental replanners reduce this latency, the planning stack remains organized around a batch path-then-time interface rather than a fixed-rate trajectory update.

Y. Cao is with Voryx Robotics LLC, San Jose, CA 95110 USA (e-mail: yongyancao@gmail.com).

This paper asks whether the timed trajectory can instead be generated directly, at the planning rate, while preserving convex kinematic constraints. The answer rests on an architectural choice rather than a property of the double integrator itself: an isolated double integrator is of course linear time-invariant, but a model predictive planner built on the *physical* plant must re-linearize and rebuild its prediction matrices at every operating point. By planning instead on a virtual double-integrator backbone, the state-transition matrix A_d is held constant across all robot configurations, so the horizon prediction matrices can be computed once offline and reused at every replanning cycle. The online problem is reduced to a convex Quadratic Program (QP) with fixed dynamics structure; only the current state, reference terms, and active constraint rows change. The contribution is a planner that, from this offline-precomputed and configuration-independent prediction structure, obtains hard kinematic feasibility and non-halting reactive replanning *together*, at the cost of one convex QP per cycle—a combination a path-then-time pipeline provides only in separate stages. Concretely, this substitutes the geometric $s \to t$ interface by a fixed-structure time-domain QP; it is not the observation that a double integrator is linear.

The idea is inspired by a two-layer predictive impedance-control architecture for physical human–robot interaction (pHRI) [3], where dynamic cancellation exposes a constant double-integrator residual. That architecture has since been verified on a physical UR5e arm [3]: with a 1.5 kg load hung during a hold, its disturbance-augmented controller stays at 0.023 mm (the encoder noise level), against 1.5 mm without the estimator, over five alternated pairs. Here the double integrator is a *planning backbone*: the planner optimizes reference trajectories for a downstream controller (computed-torque, impedance, admittance, or other), so the contribution is a configuration-independent time-domain parameterization enabling fast repeated QP updates.

The resulting framework is a **local predictive motion planner**; we refer to it throughout as the **DI-QP planner** (the double-integrator backbone solved as a receding-horizon QP), and use "time-first" only when contrasting its time-domain structure with the "path-first" TOTG pipeline. It does not perform global search; obstacle avoidance is handled by linearized safe-corridor rows or by artificial-potential-field reference deflection, both of which are local mechanisms. Within that scope, the framework provides a fixed prediction structure whose size is independent of robot configuration (growing only with horizon, dimension, and active rows);

direct optimization of time-indexed joint positions, velocities, and accelerations on a fixed grid; enforcement of velocity, acceleration, and linearized obstacle constraints inside one QP; elimination of the separate spatial-to-temporal stage; and continuous C^1 output with hard-bounded piecewise-constant acceleration, plus a piecewise-jerk extension when continuous acceleration is required.

The paper is organized as follows. Section II presents the mathematical backbone. Section III describes the receding-horizon QP formulation. Section IV details two complementary obstacle avoidance strategies. Section V analyzes key architectural properties. Section VI reports experimental benchmarks. Sections VII–IX present implementation notes, discussion, and conclusions.

II. Framework and System Structure

Let $q \in \mathbb{R}^n$ denote robot configuration and $p = f(q)$ task position. The planner generates $(q(t), \dot{q}(t), \ddot{q}(t))$ satisfying waypoint, joint, velocity, acceleration, and obstacle constraints; actuator commands are left to a downstream tracking controller. The trajectory space is parameterized by a virtual double-integrator backbone,

$$\ddot{q} = u, \qquad x = \begin{bmatrix} q^\top & \dot{q}^\top \end{bmatrix}^\top, \tag{1}$$

whose state contains planned positions and velocities. It is a configuration-independent trajectory generator that makes acceleration a decision variable and therefore keeps velocity and acceleration bounds linear.

A. Double-Integrator Backbone from Dynamic Cancellation

The same backbone can also be derived from the standard manipulator dynamics $M(q)\ddot{q} + C(q,\dot{q})\dot{q} + g(q) = \tau$. With the computed-torque (inverse-dynamics) feedforward law—the same cancellation used in the pHRI architecture of [3]—

$$\tau = M(q)u + C(q,\dot{q})\dot{q} + g(q), \tag{2}$$

substitution gives the backbone in (1), and hence

$$\dot{x} = \begin{bmatrix} 0 & I \\ 0 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ I \end{bmatrix} u. \tag{3}$$

The present work uses this residual model as a planning backbone rather than as an impedance-regulation controller.

B. Exact ZOH Discretization and the Constant A_d Property

Because the continuous system matrix is nilpotent, $A_c^2 = 0$, exact ZOH discretization over Δt gives

$$A_d = \begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix}, \qquad B_d = \begin{bmatrix} \frac{\Delta t^2}{2} \\ \Delta t \end{bmatrix}. \tag{4}$$

Unlike a physical plant linearization, this A_d is independent of configuration, operating point, and joint index. Therefore the horizon prediction matrices

$$\Phi = \begin{bmatrix} A_d \\ A_d^2 \\ \vdots \\ A_d^N \end{bmatrix}, \qquad \Gamma = \begin{bmatrix} B_d & 0 & \cdots & 0 \\ A_d B_d & B_d & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ A_d^{N-1} B_d & \cdots & A_d B_d & B_d \end{bmatrix} \tag{5}$$

are precomputed once and reused at every planning cycle. Online rollout then reduces to

$$X = \Phi x(k) + \Gamma U, \tag{6}$$

decoupling prediction cost from robot configuration.

III. Receding-Horizon Optimization

A. Optimization Problem Structure

At each planning time step, the planner solves for the optimal acceleration profile sequence across the prediction horizon:

$$U = \begin{bmatrix} u^\top(0) & u^\top(1) & \cdots & u^\top(N-1) \end{bmatrix}^\top \in \mathbb{R}^{nN}. \tag{7}$$

where n is the number of joints and N is the horizon length (e.g., $nN = 7 \times 20 = 140$ decision variables for the 7-DOF FR3 with $N = 20$). The predicted trajectory follows (6), which is affine in U; therefore kinematic bounds and linearized obstacle rows remain convex. Absent obstacle coupling, the Hessian is block-diagonal and the problem separates into n independent per-joint QPs of size N.

B. Cost Function: Tracking and Smoothness

The objective function minimizes the weighted sum of tracking deviations from a goal state $x_{\text{goal}} = [q_{\text{goal}}, 0]^\top$ and control effort. We write the stage cost with the conventional $\frac{1}{2}$ scaling,

$$J(U) = \frac{1}{2} \left(\Phi x(k) + \Gamma U - \mathbf{1}_N \otimes x_{\text{goal}}\right)^\top \bar{Q} \times \left(\Phi x(k) + \Gamma U - \mathbf{1}_N \otimes x_{\text{goal}}\right) + \frac{1}{2} U^\top \bar{R} U. \tag{8}$$

so that collecting the terms depending on U gives

$$\min_U \ \frac{1}{2} U^\top H U + h^\top U \tag{9}$$

where:

$$H = \Gamma^\top \bar{Q} \Gamma + \bar{R}, \qquad h = \Gamma^\top \bar{Q}\, x_{\text{free,err}} \tag{10}$$

and $x_{\text{free,err}} = \Phi x(k) - \mathbf{1}_N \otimes x_{\text{goal}}$ is the unforced deviation from the goal; h is the gradient at $U = 0$, i.e. the linear-cost vector passed to OSQP. The remaining definitions are: the full goal state $x_{\text{goal}} = [q_{\text{goal}}, 0]^\top$ replicated across all N horizon steps; stage weights $Q = \text{blkdiag}(K_q, D_q)$ on position/velocity error; control cost R regularizing acceleration; terminal weight $Q_f = \gamma Q$ ($\gamma \approx 5$–10); and horizon-stacked $\bar{Q} = \text{blkdiag}(Q, \ldots, Q, Q_f)$, $\bar{R} = I_N \otimes R$.

Because H is the sum of a positive semi-definite matrix (from $\bar{Q}$) and a positive definite matrix (from $\bar{R}$ with $R \succ 0$), the QP is **strictly convex** for all configurations and all time steps. A unique global minimizer exists and is found efficiently by operator-splitting solvers such as OSQP [4] with warm-starting, converging in under 0.5 ms for the decoupled box-constrained QP and in a few milliseconds for the coupled-obstacle QP (Section VI-B).

Weight tuning guidance. The weights have intuitive roles: K_q sets convergence aggressiveness (larger pulls to the goal faster but risks acceleration saturation); D_q provides damping,

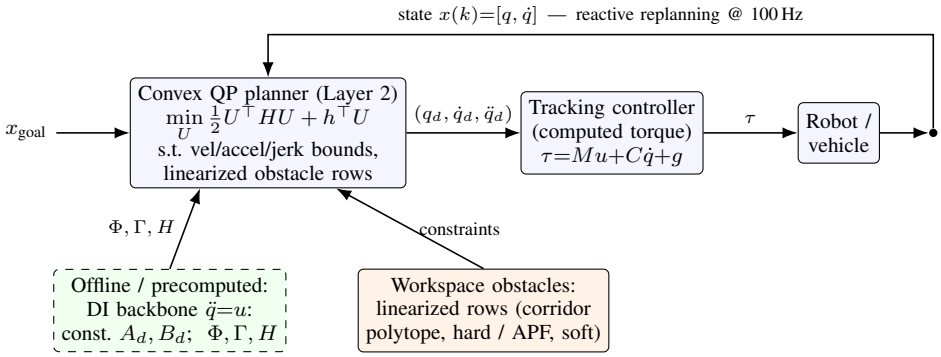


Fig. 1. Reactive predictive motion-planning architecture. The planner runs on a virtual *double-integrator backbone* ($\ddot{q} = u$), whose nilpotent dynamics give a configuration-independent constant A_d, so the horizon prediction matrices Φ, Γ and the QP Hessian H are precomputed *offline* (left, dashed). Online, a convex QP (Layer 2) optimizes the acceleration profile U subject to velocity/acceleration/jerk bounds and linearized workspace-obstacle rows (convex corridor polytope, hard; or APF goal deflection, soft), emitting a time-indexed trajectory $(q_d, \dot{q}_d, \ddot{q}_d)$ to a downstream tracking controller and plant. Current state $x(k)$ and updated obstacles re-enter every cycle, giving fixed-size reactive updates without the stop-and-rebuild of path-first pipelines; Section VI reports the measured 100 Hz timing distribution and tail.

with $D_q \approx 2\sqrt{K_q}$ giving a critically-damped response; R trades smoothness against speed; and the terminal multiplier $\gamma = 5$–10 compensates for the finite horizon at $N \geq 20$. The weights are per-joint scalars, tuned by increasing K_q until goals are reached within limits, then D_q until overshoot disappears, then R if jerk is excessive. The experiments do not share a single setting, so we list what each actually uses: the static FR3 benchmark and the Hessian-norm study run the planner defaults $K_q = 50$, $D_q = 0$, $R = 0.05\,I$, $\gamma = 8$; the FR3 dynamic-obstacle study uses $K_q = 6$, $D_q = 8$, $R = 0.05\,I$, $\gamma = 8$; the Cartesian-DI vehicle benchmark uses $K_q = 12$, $D_q = 4$, $R = 0.3\,I$, $\gamma = 6$; and the jerk-bounded lateral planner reported in Table III uses $w_q = 20$, $w_a = 0.2$, $w_r = 0.02$, $\gamma = 10$. Note that $D_q = 0$ in the static FR3 case: there the terminal multiplier and the acceleration bound alone suppress overshoot, so no velocity-error damping is needed, and the $D_q \approx 2\sqrt{K_q}$ guidance above applies to the cases that do use it rather than to every experiment reported here.

C. Kinematic Hard Constraints

Physical joint limits are enforced as hard inequality constraints on the predicted state and input sequences. For each joint i and each step k along the horizon:

$$\begin{aligned} -\dot{q}_{i,\max} \leq \dot{q}_i(k) \leq \dot{q}_{i,\max}, \qquad & k = 1, \ldots, N, \\ -u_{i,\max} \leq u_i(k) \leq u_{i,\max}, \qquad & k = 0, \ldots, N-1. \end{aligned} \tag{11}$$

Since the predicted states are affine in U by (6), the bounds in (11) translate to linear inequalities $C_{\text{kin}}U \leq d_{\text{kin}}$ assembled once from the precomputed Γ matrix and fixed online. Optionally, joint position limits $q_{i,\min} \leq q_i(k) \leq q_{i,\max}$ can be appended with no additional computational overhead.

Because the velocity bounds are imposed without slack, feasibility of the QP assumes the current state lies inside the velocity polytope; if a disturbance drives $\dot{q}(k)$ outside it, the box rows should be softened on the first horizon step or a terminal maximal-braking-to-rest set appended to recover recursive feasibility (see Section VIII-A).

IV. Incorporating Workspace Obstacle Avoidance

The linearity of the predicted joint positions with respect to U makes it straightforward to incorporate obstacle avoidance constraints without breaking convexity. Two complementary methods are presented.

A. Convex Corridor Polytope Constraints (Hard Constraints)

Cartesian obstacles detected by perception systems are represented as half-space inequalities in task space:

$$A_{\text{obs}}\, p(k) \leq b_{\text{obs}} \tag{12}$$

where $p(k) \in \mathbb{R}^3$ is the predicted end-effector (or link) position at step k. Using the robot's forward kinematics linearized at the current operating point via the translational Jacobian $J_v(q_0)$:

$$p(k) \approx p_0 + J_v(q_0)\big(q(k) - q_0\big) \tag{13}$$

the workspace constraint in (12) becomes:

$$A_{\text{obs}} J_v(q_0)\, \Delta q(k) \leq b_{\text{obs}} - A_{\text{obs}} p_0 \tag{14}$$

where $\Delta q(k) = q(k) - q_0$ is linear in U through Γ. To make the QP row explicit, let $E_{q,k} \in \mathbb{R}^{n \times 2nN}$ be the selector that extracts the n-dimensional joint-position block of the stacked state X at horizon step k (a row of $n \times n$ identity blocks, zero elsewhere), so

$$q(k) = E_{q,k}\big(\Phi x(k_0) + \Gamma U\big). \tag{15}$$

Define $C_{\text{obs},k} = A_{\text{obs}} J_v(q_0) E_{q,k} \Gamma$. Then each obstacle row has the affine form

$$\begin{aligned} C_{\text{obs},k} U \leq\; & b_{\text{obs}} - A_{\text{obs}} p_0 \\ & - A_{\text{obs}} J_v(q_0)\big(E_{q,k}\Phi x(k_0) - q_0\big) =: d_{\text{obs},k}. \end{aligned} \tag{16}$$

Thus the free-response term is absorbed into the right-hand side, and the obstacle constraint becomes a set of affine inequalities in U, fully compatible with the convex QP structure.

Approximation quality. Because $J_v(q_0)$ is held constant over the horizon, these rows are a first-order local approximation of the safe corridor—exact near q_0, increasingly inaccurate away from it, and not automatically conservative. Relinearizing every cycle refreshes it; a conservative half-space

additionally tightens each row by a bound on the linearization error. From the second-order Taylor remainder, with $H_p(q_0)$ the position Hessian, the true end-effector position satisfies

$$\|p(q) - (p_0 + J_v(q_0)\Delta q)\| \le \tfrac{1}{2}\|H_p(q_0)\|_F \|\Delta q\|^2 \quad (17)$$

where $\|\cdot\|_F$ is the Frobenius norm. For an obstacle half-space row $a^\top p \le b$, a conservative tightened row is

$$a^\top \big(p_0 + J_v(q_0)\Delta q(k)\big) \le b - \|a\|\,\varepsilon_p, \quad (18)$$

with $\varepsilon_p \ge \frac{1}{2}\|H_p\|_F\|\Delta q\|^2$ from (17). For the FR3, $\|H_p\|_F$ sampled at 5000 joint-range configurations has mean 1.64 and 95th-percentile 1.96 m/rad² (`fr3_hessian_norm.py`). The measured end-horizon error grows quadratically with joint-speed×horizon (harness: 7.1 mm at 25% speed over 0.1 s, 106.4 mm at 50% over 0.2 s, 1.03 m at full speed over 0.4 s). The Frobenius bound is a worst case over directions and is therefore loose: at 50% speed over the $N = 20$ horizon $\|\Delta q\|_2 \approx 1.0$ rad, so (18) prescribes $\varepsilon_p \gtrsim \frac{1}{2}(1.96)(1.0)^2 \approx 1.0$ m, whereas the realized end-horizon drift is only 106 mm because the executed motion does not excite the worst-case curvature. In practice the margin should therefore be set from the *measured* drift—a workspace percentile of the end-horizon error—rather than from the conservative Frobenius bound, and the horizon shortened for fast motions; at modest speeds a centimetre-scale margin suffices.

Accuracy of the constraint guarantee. The QP guarantees satisfaction of the *linearized* safe-corridor constraints: no predicted trajectory penetrates the declared linearized obstacle polytope. A true nonlinear collision-clearance guarantee additionally requires conservative tightening by ε_p, nonlinear collision checking of the accepted rollout, or both; for non-convex environments, standard safe-corridor decomposition [5] can be applied upstream.

Multiple obstacles. Method A appends M independent half-space sets, keeping convexity at mildly increasing solver cost; if rows conflict the QP becomes infeasible, resolved by per-obstacle slacks $s_m \ge 0$ with penalty $\mu\|s\|^2$. Method B sums individual APF repulsion fields, avoiding infeasibility but inheriting the local-minimum risk of scalar APF methods.

B. Artificial Potential Field Target Deflection (Soft Constraint)

For scenarios requiring maximum solver throughput, an alternative soft-constraint approach using Artificial Potential Fields (APF) [6] is available. An obstacle at position p_{obs} with influence radius ρ_0 and repulsive gain η generates a task-space repulsive potential:

$$\mathcal{U}_{\text{rep}}(p) = \begin{cases} \frac{1}{2}\eta\left(\frac{1}{\|p-p_{\text{obs}}\|} - \frac{1}{\rho_0}\right)^2, & \|p - p_{\text{obs}}\| \le \rho_0, \\ 0, & \text{otherwise.} \end{cases} \quad (19)$$

The negative gradient $g_{\text{rep}}(p) = -\nabla\mathcal{U}_{\text{rep}}(p)$, treated as a synthetic direction field rather than a physical force, is normalized as $\hat{g}_{\text{rep}} = g_{\text{rep}}/(\|g_{\text{rep}}\|+\epsilon)$ and mapped to a heuristic joint-space target shift via the Jacobian pseudo-inverse,

$$\delta q_{\text{obs}} = K\, J_v^+(q)\, \hat{g}_{\text{rep}}, \quad (20)$$

where the scalar gain $K > 0$ absorbs all dimensional scaling. (In the simulations below the same idea is implemented as a velocity/reference deflection with an optional tangential component to avoid stalls; in both cases APF changes the reference rather than adding a hard row.) The target fed to the cost function is then updated online:

$$q_{\text{target}}(k) = q_{\text{goal}} + \delta q_{\text{obs}}. \quad (21)$$

This bypasses inequality rows entirely, preserving the minimal QP structure and maximizing solve speed. The trade-off is that obstacle avoidance is enforced softly through the cost function rather than guaranteed through hard constraints; APF-based avoidance is also susceptible to local minima in complex environments (see Section VIII).

Local minima mitigation. When the APF gradient vanishes the planner stalls; three escape strategies are available: (1) *waypoint injection*—insert a lateral waypoint after a no-progress timeout (e.g. 0.5 s); (2) *random perturbation*—add a small random shift to q_{target} for a few cycles; (3) *hybrid switch*—activate the Method-A hard polytope (with slack) on stall. Option (1) is preferred as it adds no solver overhead. The two methods are complementary: hard polytope constraints for static infrastructure, APF deflection for dynamic obstacles.

V. Architectural Properties

A. Constant-Structure MPC and Fixed-Size Replanning Updates

The central property is that all dynamics prediction matrices (Φ, Γ, H) are precomputed offline and never rebuilt online; obstacle rows are re-assembled each cycle but still project J_v through the fixed Γ. Each cycle thus executes a QP whose dynamics structure is fixed at initialization. Per-cycle work consists of one QP update and solve for a given active set—growing with the obstacle-row count but independent of configuration and free of prediction-matrix rebuilds. This is distinct from configuration-dependent MPC (which recomputes prediction matrices at each linearization point) and from batch planning (TOTG, TOPP-RA, where any workspace change triggers a full rebuild). The constant-A_d property makes the update size predictable; it does not by itself establish a hard execution-time bound, so Section VI reports p95 and maximum latency separately.

B. Elimination of Spatial-to-Temporal Transformations

Traditional pipelines solve a separate time-parameterization problem—given a path $\sigma(s)$, find $s(t)$ satisfying velocity/acceleration limits—decoupled from obstacle avoidance and re-executed whenever the path changes. The proposed framework has no geometric path: positions, velocities, and accelerations along the horizon are generated simultaneously in one QP, so the $s \leftrightarrow t$ conversion stage and its singularities (Section VI.A) are eliminated by construction.

C. C^1 Output with Hard-Bounded Acceleration

Because the proposed planner uses joint acceleration u_i as the direct optimization variable and enforces acceleration bounds as hard constraints, the output $(q_d, \dot{q}_d)$ is C^1 with $\ddot{q}_d$ hard-bounded at all times. Since u_i is held over each control

cycle (zero-order hold), $\ddot{q}_d$ is piecewise-constant and steps between cycles; full C^2 continuity requires jerk to be included as the optimization variable within a higher-order planning backbone.

D. Piecewise-Jerk Triple-Integrator Extension

When continuous acceleration is required—to reduce torque-rate excitation, satisfy comfort limits, or bound steering-rate in vehicle applications—the same constant-A_d principle extends to a triple-integrator backbone that optimizes jerk $\dddot{q}_i = j_i$ on the state $z_i = [q_i, \dot{q}_i, \ddot{q}_i]^\top$. The continuous system matrix is again nilpotent ($A_c^3 = 0$), so exact ZOH discretization gives the configuration-independent

$$A_d^{(3)} = \begin{bmatrix} 1 & \Delta t & \frac{1}{2}\Delta t^2 \\ 0 & 1 & \Delta t \\ 0 & 0 & 1 \end{bmatrix}, \qquad B_d^{(3)} = \begin{bmatrix} \frac{1}{6}\Delta t^3 \\ \frac{1}{2}\Delta t^2 \\ \Delta t \end{bmatrix}. \tag{22}$$

so the prediction matrices remain precomputable offline. With jerk held constant per interval, $\ddot{q}(t+\tau) = \ddot{q}(t) + \tau j(t)$ keeps acceleration continuous across samples while jerk is piecewise-constant, raising output smoothness from C^1 to C^2 ($q_d \in C^2$, $\dot{q}_d \in C^1$, $\ddot{q}_d \in C^0$) with hard linear bounds now available on velocity, acceleration, and jerk.

The trade-off is a larger state and extra predicted-state rows, though the decision sequence is still length nN, now a jerk sequence $J = [j(0), \ldots, j(N-1)]$ rather than acceleration. The AV steering-rate benchmark (§VI.C) uses exactly this for the lateral channel, since steering rate depends on lateral jerk.

Its objective differs from (8) and is stated here explicitly. Writing Γ_p, Γ_a for the position and acceleration blocks of the triple-integrator prediction matrices, $e_p = \Phi_p z - y_{\text{goal}}$ for the free-response lateral offset error and $a_{\text{free}} = \Phi_a z$ for the free-response lateral acceleration, the lateral planner minimizes

$$\begin{aligned} J(U) = {} & \tfrac{1}{2}\left(\Gamma_p U + e_p\right)^\top \bar{W}_q \left(\Gamma_p U + e_p\right) \\ & + \tfrac{1}{2} w_a \left\|\Gamma_a U + a_{\text{free}}\right\|^2 + \tfrac{1}{2} w_r \|U\|^2, \end{aligned} \tag{23}$$

with $\bar{W}_q = \operatorname{diag}(w_q, \ldots, w_q, \gamma w_q)$ and $w_q = 20$, $w_a = 0.2$, $w_r = 0.02$, $\gamma = 10$. Two differences from (8) matter: the middle term penalizes the predicted *absolute* lateral acceleration rather than a velocity error, because lateral acceleration is the comfort-relevant quantity here; and there is no velocity-error term at all, since the lateral channel tracks an offset rather than a velocity profile. Velocity, acceleration, and jerk remain hard rows in the constraint set. The jerk row bounds lateral jerk, not steering rate itself: because $\delta = \arctan\big(L(\dot{x}\ddot{y} - \dot{y}\ddot{x})/(\dot{x}^2 + \dot{y}^2)^{3/2}\big)$ depends nonlinearly on the lateral velocity and acceleration as well, a given $j_{\max}$ implies a steering-rate cap only once those terms are accounted for at the operating point. The harness therefore *measures* the resulting steering rate (§VI.C) rather than constraining it in the QP.

The rest of the paper uses the double-integrator backbone as the main manipulator planner because it is the smallest model that enforces velocity and acceleration limits directly and gives the fastest QP. The triple-integrator form is a drop-in extension when continuous acceleration or explicit jerk bounds are application-critical.

E. Analytical LQR Characterization of the Double-Integrator Backbone

As background, the unconstrained infinite-horizon limit of the planner is the standard LQR law $u = u_{ff} - K(x - x_r)$, $K = R^{-1}B^\top P$, with P solving the Algebraic Riccati Equation $A^\top P + PA - PBR^{-1}B^\top P + Q = 0$ ($u_{ff} = \ddot{q}_r$, zero for fixed-goal regulation). This situates the method within standard optimal control and relates the weights to the closed-loop second-order behavior. The constrained planner is the finite-horizon counterpart: the same backbone solved as a receding-horizon QP enforcing velocity, acceleration, joint, and obstacle constraints—mathematically close to linear MPC [9], but on a virtual trajectory state rather than a plant linearization. The robot model enters through kinematics and workspace constraints [8]; the output is the reference trajectory $\mathcal{T} = \{q_d, \dot{q}_d, \ddot{q}_d\}$ for a downstream controller.

VI. Experimental Evaluation

We evaluate the proposed planner against a path-first Time-Optimal Trajectory Generation pipeline on a 7-DOF Franka FR3 manipulator and on an autonomous-vehicle steering task. Both planners receive identical waypoints and kinematic limits and emit $(q_d, \dot{q}_d, \ddot{q}_d)$ sampled at the same Δt; all numbers reported below are measured by the accompanying open harness: the scripts in `benchmarks/`, which print each reported table to stdout, with the rendered figures and the AV trajectory export collected in `simulationResults/`. Because several benchmark modules re-export a shared planner from a sibling repository, a frozen snapshot with those modules inlined is shipped as `benchmarks_standalone.zip`, together with source provenance and a dependency manifest. All non-MuJoCo tables reproduce from that archive alone. The computed-torque execution table additionally requires the third-party Franka model/assets via `FR3_XML`; those assets are not bundled. Raw per-trial dumps are not committed—the harness recomputes them—and wall-clock values are hardware/OS dependent, so deterministic constraint and geometry metrics are distinguished from timed measurements. The TOTG reference uses the numerical-integration TOPP core—circular-blend path (Kunz–Stilman geometry [10]), time-optimal forward/backward sweep on $\dot{s}^2$ [7], then $s \to t$ inversion and resampling.

Scope of comparison. We compare against the path-first family (TOTG, TOPP-RA), a resolved-rate-plus-APF baseline, and a sampling-based MPPI baseline on the planar dynamic-obstacle test (Section VI-D). Online nonlinear MPC and CHOMP/TrajOpt are deferred to future work; the present planner's appeal is the constant-A_d precomputation and fixed-size online update, while richer nonlinear formulations address a broader planning model.

This section reports four evaluation axes: reactive latency, jerk/ smoothness, constraint fidelity, and the execution-time gap. On static paths, time-optimal path parameterization is faster; the fixed time-domain QP trades a small amount of that global time optimality for fixed-size reactive updates, explicit limit satisfaction, and smoother command profiles.

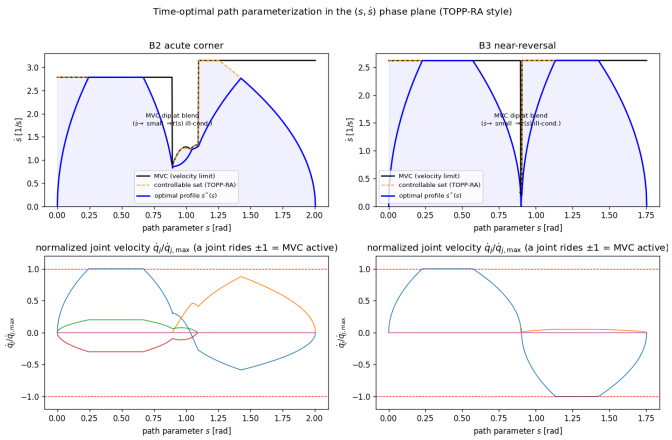


Fig. 2. Time-optimal path parameterization in the $(s, \dot{s})$ phase plane (TOPP-RA style). The optimal profile dips at the tight blend and—on the near-reversal—plunges to $\dot{s} \approx 0$, exactly where the $s \to t$ map becomes singular.

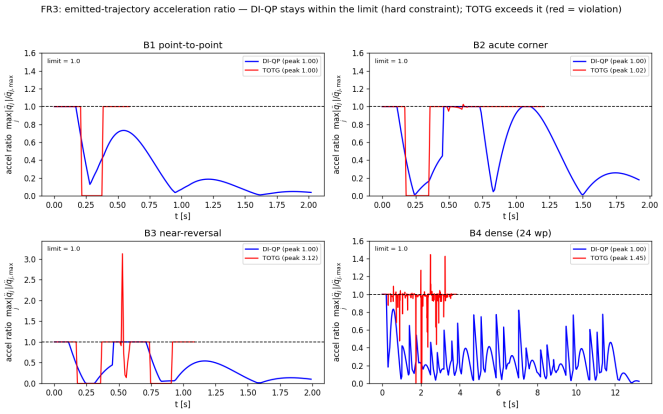


Fig. 3. FR3 emitted-trajectory acceleration ratio $\max_j |\ddot{q}_j|/\ddot{q}_{j,\max}$ over time, DI-QP (blue) vs TOTG (red), B1–B4; dashed line is the limit. DI-QP is pinned within the limit by hard constraints; TOTG exceeds it at blend seams (B2 $1.047\times$, B3 up to $3.13\times$, B4 $1.445\times$). The B3 spike is blend-dependent (drops to $1.00\times$ under TOPP-RA/C^2); the intrinsic path-first issues are the $\dot{s} \to 0$ singularity and curvature-discontinuous reconstruction, not the spike magnitude.

A. *Path-First $t \leftrightarrow s$ Conversion Artifacts*

The $s \to t$ reconstruction differentiates the geometric path,

$$\ddot{q}(t) = q'(s)\,\ddot{s} + \underbrace{q''(s)}_{\text{curvature vector}}\,\dot{s}^2.$$

Two pathologies follow from the standard circular-blend path representation. First, $q''(s)$ jumps at every blend seam (0 on a straight segment, $1/r$ on a circular arc), so the reconstructed $\ddot{q}$ is discontinuous, with the jump scaling as $\dot{s}^2$. Second, $t(s) = \int ds/\dot{s}$ is singular wherever $\dot{s} \to 0$ (rest points, tight near-reversal blends), making the uniform-time resample ill-conditioned. These artifacts are a property of the circular-blend path representation, not of time parameterization per se. Replacing circular blends with a C^2 path (e.g., cubic splines) removes them, as confirmed below. The time-first planner never forms s; its $\ddot{q}$ is the bounded decision variable, so neither pathology can arise.

The $(s, \dot{s})$ phase plane (Fig. 2) makes the singularity visible: the maximum-velocity curve and optimal profile $\dot{s}^*(s)$ dip toward zero at the tight blend—severely on the near-reversal, where $\dot{s} \to 0$ drives $t(s)$ singular.

B. *FR3 Manipulator Results*

Stresses are concentrated in two joints for interpretability while all seven are planned. The decisive metric is the peak acceleration ratio $\|\ddot{q}\|_\infty / \ddot{q}_{\max}$: a value above 1 means the emitted trajectory violates the declared acceleration limit. The DI planner enforces this as a hard QP constraint and is pinned at 1.000 with zero violations in every scenario.

Separating the decoupled and coupled cases, the single-threaded Python harness (OSQP tolerance 10^{-6}, warm-started, fixed KKT sparsity on the coupled update path) solves the 140-variable decoupled box-constrained QP (7 DOF, N=20) in 0.03–0.08 ms mean on the current workstation. Across three repeated fixed-sparsity runs, the 141-variable coupled obstacle QP (Method A) has solve mean/p95/max ranges of 1.87–2.07/3.87–4.49/8.34–9.19 ms; including Python assembly and updates, the full-cycle ranges are 4.74–5.07/6.60–7.70/12.26–18.48 ms.

A horizon sweep $N = \{10, 20, 30, 40\}$ on the same coupled FR3 dynamic-obstacle problem (`sweep_N.py`) confirms why $N = 20$ (0.2 s) is the default. Because the scenario uses a velocity-law/APF reference, completion (1.83–1.86 s) and min EE clearance (≈0.30 m) are nearly insensitive to N; the dominant trend is computational, with coupled-QP solve p95 rising steeply from 1.0 ms (N=10) and 4.4 ms (N=20) to 20.4 ms (N=30) and 55.6 ms (N=40). For pure position-tracking waypoints $N = 10$ instead inflates time-to-goal, so $N = 20$ balances both regimes.

Comparison scope. The acceleration violation metric compares the trajectory as emitted by each planner—i.e., the command sequence as-sent to the controller, with no downstream smoothing or re-interpolation applied to either planner's output. In practice, some systems apply a final spline pass to the TOTG output before execution, which would reduce or eliminate the resampling-induced violations on B2 and B4; under such post-processing the DI planner's zero-violation advantage on those scenarios shrinks. The advantage on B3 (near-reversal singularity) and in the reactive scenarios (Sections VI.D–F) is independent of downstream smoothing.

Figure 3 plots the acceleration ratio over time for each scenario: the DI curve never crosses 1, while the TOTG curve spikes above it at blend seams.

In **B3 (near-reversal)** the blend radius collapses, $\dot{s} \to 0$, and the parameterization becomes singular: $\max(1/\dot{s}) \to \infty$ and the $s \to t$ conversion fails (flagged and capped by the harness). In **B4** the path-first command breaks the acceleration limit by 45% at 49 samples. The DI planner remains feasible and smooth throughout.

Robustness to the parameterizer (TOPP-RA). Re-running with reachability-based **TOPP-RA** [7]—designed to be robust at dynamic singularities—the structural findings persist: the acute corner still exceeds the acceleration limit ($1.020\times$ vs $1.025\times$), the dense path still violates ($1.12\times$), and the near-

TABLE I
FR3 MANIPULATOR BENCHMARK RESULTS

Scenario	DI T [s]	TOTG T [s]	DI accel ratio	TOTG accel ratio	TOTG accel viol.	DI jerk RMS	TOTG jerk RMS
B1 point-to-point	2.02	**0.59**	1.000	1.000	0	**22**	143
B2 acute corner (tight blend)	1.92	**1.23**	1.000	**1.047**	2	36	133
B3 near-reversal	1.99	1.07	1.000	up to **3.13**	0–1	29	166
B4 dense (24 waypoints)	13.24	**3.83**	1.000	**1.445**	49	**54**	333

TABLE II
VELOCITY-LAW RECOVERY OF THE TIME-OPTIMALITY GAP (FR3)

Scenario	Planner	T [s]	vs TOTG	Jerk RMS
B1	baseline DI	2.02	3.42×	**22.4**
B1	velocity-law DI	0.78	1.32×	139.9
B1	TOTG	0.59	1.00×	143.0
B2	baseline DI	1.92	1.59×	**36.2**
B2	velocity-law DI	1.28	1.06×	293.2
B2	TOTG	1.21	1.00×	137.3
B4	baseline DI	13.24	3.46×	**53.6**
B4	velocity-law DI	4.38	1.14×	841.5
B4	TOTG	3.83	1.00×	332.9

reversal stays singular ($\min \dot{s} = 0$) for both parameterizers. TOPP-RA softens magnitudes (the $3.13\times$ B3 spike drops to $1.000\times$), separating what is implementation-dependent (spike size) from what is intrinsic to path-first (the violation from curvature-discontinuous reconstruction and the $\dot{s} \to 0$ singularity). Replacing circular blends with a C^2 spline removes the B2/B4 violations entirely, so TOPP-RA on a C^2 path matches the DI planner on those metrics.

Time-optimality. TOTG is the lower bound by construction. With the smooth quadratic-cost setting, the DI completion-time ratios are $3.4\times$ in B1 and $3.5\times$ in B4. Supplying the analytic time-optimal velocity-law reference (velocity-dominant weights, `improve_test.py`) recovers the gap to 1.06–$1.32\times$ while the DI-QP still preserves zero limit violations, as quantified in Table II. The recovery trades smoothness for speed, raising jerk RMS by roughly an order of magnitude: the velocity-law reference is an explicit speed-versus-smoothness selector. Thus the DI planner remains feasible by construction with respect to the declared time-domain bounds, while path-first feasibility can be lost after circular-blend reconstruction and uniform-time resampling unless a smoother path representation or downstream post-processing is used.

All DI rows hold zero declared-limit violations; the velocity-law rows close most of the time gap while the baseline DI rows keep the lowest jerk. Jerk RMS is in rad/s^3. The B2 corner here uses a moderate blend (max_dev $= 0.05$, TOTG $T = 1.21$ s); the tight-blend B2 of Table I (max_dev $= 0.02$, $T = 1.23$ s) is a slightly different operating point, which accounts for the small difference in the TOTG reference time.

C. Vehicle Steering-Rate Smoothness via the Lateral Backbone

The same planning/control separation appears in autonomous-driving stacks such as Apollo: planners output $\mathcal{T} = \{x, y, \theta, \kappa, v, a\}$, while downstream controllers compute steering, throttle, and braking. A virtual Cartesian double-integrator, $\ddot{p} = u$, can therefore serve as the configuration-independent planning backbone, after which heading and curvature are recovered as

$$\theta = \operatorname{atan2}(\dot{y}, \dot{x}), \quad \kappa = \frac{\dot{x}\ddot{y} - \dot{y}\ddot{x}}{(\dot{x}^2 + \dot{y}^2)^{3/2}}. \tag{24}$$

Vehicle feasibility then requires speed, acceleration, curvature, and steering-rate constraints or post-checks on top of the parameterization. The AV benchmark evaluates one concrete claim: that applying the constant-A_d principle to the lateral channel—optimizing lateral jerk, the derivative that sets steering rate, then recovering vehicle commands—produces a steering profile that stays within a representative actuator-rate envelope where the path-first circular-blend command does not. The result is a lateral-channel claim at a fixed cruise speed, complete and tested for that channel.

With a representative steering-rate cap of 0.5 rad/s, the jerk-bounded time-first planner stays within the actuator-rate envelope in both cruise maneuvers, whereas the path-first command exceeds the cap on the double lane change because circular-blend curvature steps at path seams. Both maneuvers hold a fixed cruise speed of 13 m/s; since steering rate scales with forward speed for a fixed path curvature profile ($\dot{\delta} \propto \dot{\kappa} v + \kappa \dot{v}$), this margin is specific to that operating point. The pure Cartesian double-integrator variant, also reported by the harness, satisfies steering-rate constraints once curvature/rate constraints or a higher-order lateral state are added.

D. Fixed-Size Reactive Replanning

The reactivity advantage is structural rather than implementation-dependent. A planar point robot must reach a goal while a disk obstacle crosses its straight-line path mid-execution. The DI planner recomputes an APF velocity reference from the obstacle's current position each 100 Hz cycle and re-solves the box-constrained QP, deflecting continuously. The path-first planner must halt, rebuild a detour path, and re-run TOPP from rest.

Both reach the goal collision-free, but with different cost structures. On an Intel Xeon w7-2475X under Windows 11, Python 3.12.12, and OSQP 1.1.3, three repeated runs of this fixed-size APF benchmark give DI mean cycle latency of 0.118–0.130 ms and worst reaction latency of 1.92–2.22 ms. The corresponding path-first rebuild takes 26.9–40.7 ms before any added search delay. We sweep an additional path-search latency from 0 to 400 ms:

TABLE III
AUTONOMOUS-VEHICLE STEERING BENCHMARK

Scenario	Planner	Peak steering [deg]	Peak steering rate [rad/s]	Steering jump [rad]	Peak lateral accel [m/s²]
Single lane change	jerk-bounded time-first	3.6	**0.130**	**0.0065**	3.99
Single lane change	TOTG path-first	1.5	0.254	0.0254	2.47
Double lane change	jerk-bounded time-first	3.6	**0.130**	**0.0065**	3.99
Double lane change	TOTG path-first	3.4	0.594	0.0594	4.05

TABLE IV
DYNAMIC-OBSTACLE REPLANNING LATENCY SWEEP

Added path-search latency	TOTG completion [s]	TOTG time halted [s]	DI completion [s]
0 (TOPP only)	6.31–6.33	0.12–0.14	**6.53 (flat)**
100 ms	6.38–6.40	0.39–0.41	6.53
200 ms	6.66–6.68	0.70–0.72	6.53
400 ms	7.20–7.21	1.29–1.30	6.53

TABLE V
REACTIVE BASELINES ON THE PLANAR DYNAMIC-OBSTACLE TEST

Metric	DI-QP	MPPI
Reached goal	yes	yes
Completion time [s]	6.53	6.12
Min clearance [m]	0.485	0.483
Velocity limit enforced by	QP constraint	clamping
Cycles needing velocity clamp	**0**	166
Rollouts per cycle	1 QP	512×20
Mean cycle latency [ms]	**0.118–0.130**	1.886–1.979
Worst cycle latency [ms]	**1.92–2.22**	8.31–9.97

The structural advantage is bounded by a fixed optimization structure whose size grows with the number of active obstacle constraints: DI deflects within one cycle and never halts, whereas TOTG must stop and rebuild. The reactive edge is decisive when replanning is expensive (full geometric search) or obstacles are fast, and marginal when the rebuild is cheap: with zero replan latency, TOTG and DI have comparable completion times.

Comparison with a sampling-based reactive baseline. The stop-and-rebuild TOTG baseline isolates the reactivity gap against path-first pipelines, but a fair question is how the planner compares to a reactive method that, like ours, never halts. We therefore add a Model-Predictive Path Integral (MPPI) [15] controller on the *same* planar double integrator and moving obstacle: each 100 Hz cycle samples $K = 512$ acceleration rollouts over an $H = 20$ horizon, scores them by a goal/obstacle/effort cost, and applies the softmax-weighted mean. Tuned to match the DI planner's clearance ($\lambda = 60, \sigma = 4$, the tightest-clearance safe setting in a temperature sweep; lower λ becomes winner-take-all and does not reach the goal in this scenario) (`benchmark_dynamic.py`), both reach the goal with essentially identical safety, but the cost structure differs sharply (Table V): MPPI shares the continuous-update property, yet enforces the velocity limit only by clamping the applied command—on 166 cycles—whereas the DI-QP enforces it as a convex constraint and never clamps. MPPI evaluates $K \times H$ rollouts per cycle; on the test workstation it uses approximately 15–16$\times$ the DI-QP mean cycle time and about 4–5$\times$ its tail-sensitive worst-cycle time. Continuous reactivity is thus shared with MPPI; obtaining it together with hard limit feasibility at QP-level cost is what the DI-QP adds. The same holds for GPU sampling MPC: STORM [16] reaches a comparable rate (< 8 ms) on a Panda but enforces joint/collision limits as soft costs; the CPU MPPI baseline isolates the constraint-handling difference without a CPU-versus-GPU hardware confound.

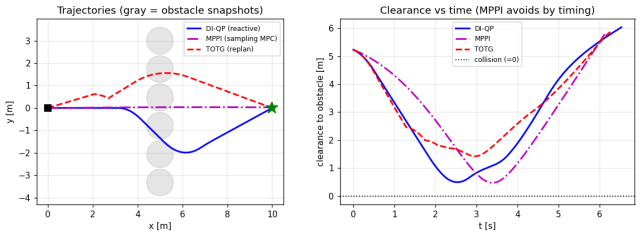


Fig. 4. Planar dynamic-obstacle test. Left: trajectories—DI-QP (solid) detours spatially, MPPI (dash-dot) avoids by *timing* its passage, path-first (dashed) halts and rebuilds; gray disks are obstacle snapshots. Right: clearance vs time, all three collision-free. DI-QP reacts without a path rebuild; across three workstation runs its mean cycle latency is 0.118–0.130 ms versus 1.886–1.979 ms for MPPI, which also lacks hard limits (Table V).

E. APF Local-Minimum Failure and Waypoint-Injection Recovery

The APF method is intentionally presented as a soft, high-throughput heuristic rather than a collision-certificate method. To expose its failure mode, we place a planar double-integrator robot, goal, and circular obstacle on the same line. With a purely radial APF field, the attractive and repulsive terms balance in front of the obstacle and the planner stalls. A simple waypoint-injection rule detects lack of progress and inserts a lateral detour sequence around the obstacle; the same box-constrained QP then reaches the goal while preserving velocity and acceleration bounds (`apf_local_minimum.py`). Pure APF stalls 6.07 m from the goal (final time 17.99 s), whereas waypoint injection reaches it (0.12 m residual, 15.56 s) at the same 0.22 m obstacle clearance and within the same 1.50 m/s and 3.00 m/s² limits, at negligible extra solve cost (0.135 vs 0.092 ms mean).

This benchmark motivates the hybrid recommendation used throughout the paper: APF reference deflection is useful for fast reactive steering, but hard polytope constraints or explicit waypoint/safe-corridor logic should be activated when

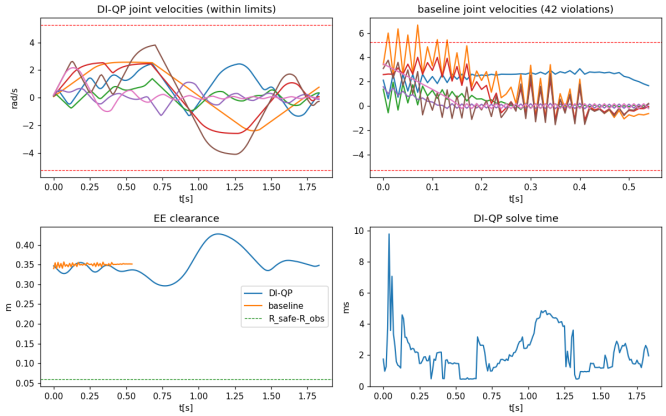


Fig. 5. 7-DOF FR3 dynamic obstacle. Top-left: DI-QP joint velocities, all within the limit. Top-right: resolved-rate+APF baseline, which exceeds the velocity limit. Bottom-left: end-effector clearance. Bottom-right: DI-QP per-cycle solve time.

progress stalls or clearance must be certified.

F. Scaling to the 7-DOF Arm with a Dynamic Obstacle

We exercise the full Section IV machinery on the 7-DOF FR3, where obstacle constraints couple the joints through the translational Jacobian. The arm reaches a joint-space goal while a spherical obstacle crosses the end-effector path. We compare against a resolved-rate-plus-APF reactive controller without hard limits.

Both avoid the obstacle and reach the goal. The reactive baseline—lacking hard constraints—exceeds the joint velocity limit at 42 samples and acceleration at 346, whereas the QP holds every joint limit exactly and keeps the linearized half-space feasible (zero slack), at a min EE clearance of 0.296 m versus 0.339 m for the unconstrained baseline. The fixed-sparsity OSQP update path (`fixed_sparsity_timing.py`) gives the repeated solve and full-cycle ranges reported in Section VI-B. The single-thread Python harness meets 100 Hz at p95; full-cycle maxima of 12.26–18.48 ms identify the tail-latency target for a compiled implementation with reusable active-set or factorization state.

G. Randomized FR3 Dynamic-Obstacle Robustness

To check that the dynamic-obstacle result is not tied to a single hand-picked configuration, we perturb the start and goal joint states, obstacle lateral offset, initial height, and crossing speed over 20 randomized trials (`fr3_dynamic_randomized.py`, seed 7). Both planners reach the goal in all trials and avoid the obstacle. The distinction is constraint fidelity: the QP maintains hard joint limits in every trial, while the resolved-rate baseline repeatedly exceeds them.

The baseline tends to keep slightly larger clearance because it is unconstrained and can command arbitrarily aggressive joint velocities and accelerations. The QP accepts a smaller but still positive clearance while preserving every declared joint limit.

TABLE VI
RANDOMIZED FR3 DYNAMIC-OBSTACLE ROBUSTNESS

Metric	DI-QP	Reactive baseline
Success rate	100%	100%
Min EE clearance, mean ± std [m]	0.293 ± 0.018	0.328 ± 0.021
Worst min EE clearance [m]	0.261	0.266
Total joint velocity violations	**0**	618
Total joint acceleration violations	**0**	6889
Max slack used [m]	0.0000	—
QP solve p95 / max [ms]	6.15–6.66 / 25.96–33.10	—

TABLE VII
MUJOCO COMPUTED-TORQUE EXECUTION

Metric (joint 1 / worst)	DI-QP	TOTG
Peak demanded torque [Nm]	58.0	56.7
Torque-rate RMS [Nm/s]	**181**	383
Samples over torque limit	0	0
Tracking RMSE [mrad]	1.9	2.0

The timing measurements cover different scopes. The repeated fixed-sparsity study gives 8.34–9.19 ms solve maxima and 12.26–18.48 ms full-cycle maxima; across three runs of the 20-trial randomized study, solve p95 is 6.15–6.66 ms and the maximum is 25.96–33.10 ms. The p95 remains below 10 ms, while the full-cycle and randomized maxima locate the tail that a hard-deadline implementation must reduce.

H. Closed-Loop Dynamics Simulation in MuJoCo

To verify that planner-level differences survive contact with rigid-body dynamics, we execute both references on a torque-controlled FR3 in MuJoCo (3.8.1; full mass matrix and gravity). A 500 Hz computed-torque controller tracks each 100 Hz reference on the acute-corner maneuver.

Both stay within the torque envelope and track to ≈ 2 mrad RMS. TOTG's acceleration discontinuity at the blend seam manifests in simulation as a stepped torque command (torque-rate RMS $2.1\times$ that of the DI reference), while the DI reference produces a smooth torque profile. TOTG completes the fixed path faster; the DI command is gentler on the drivetrain—the smoothness-versus-time trade-off in closed-loop dynamics rather than at the planner output.

VII. IMPLEMENTATION NOTES

The QP is solved with OSQP [4]. **Warm-starting** from the previous solution U^* cuts cold-start latency from $\sim$5 ms to under 0.5 ms for the decoupled box-constrained QPs. **Sparse structure** is reused: the Hessian H is assembled once offline, the decoupled OSQP object is set up once with only vectors updated online, and the coupled-obstacle half-space uses a fixed sparsity pattern updated in place via OSQP's `Ax` path, avoiding symbolic re-factorization each cycle. For **horizon length**, $N = 10$ (0.1 s) is too short for pure position-tracking waypoints, so the default is $N = 20$ (0.2 s); it recovers most achievable speed while keeping the coupled QP within a few milliseconds p95, whereas $N \approx 30$ narrows the static time-optimality gap only at higher jerk and solve time.

For multi-waypoint tasks the goal x_{goal} is advanced online as waypoints are reached (within ϵ_q); the receding-horizon time-domain formulation makes these transitions smooth with no path splicing or re-initialization. The online cost is dominated by the QP solve itself ($nN = 140$ variables for the 7-DOF N=20 case, 141 with a shared obstacle slack), since the precomputed Φ, Γ are formed once and never rebuilt—the practical payoff of the constant-A_d property.

VIII. Discussion

The proposed framework unifies trajectory generation and time-domain constraint enforcement in a single continuously-running optimization loop, rather than planning in space and then parameterizing in time. The constant-A_d property—from the nilpotency of the double-integrator system matrix—is the enabling insight: it decouples trajectory-prediction cost from configuration. On the test workstation, the Python implementation achieves sub-10 ms full-cycle p95, while the reported maxima remain above a hard 100 Hz deadline.

A. Operating Envelope and Validation Scope

The evaluated method is a **local** predictive planner whose operating envelope has four design conditions. **(i) Timing objective:** a pure position-tracking cost regulates like a damped second-order system, with time-to-goal of 3–3.5× TOTG; the analytic velocity-law reference recovers this to 1.06–1.32× at zero violations, acting as a speed-versus-smoothness selector. **(ii) Local linearization regime:** Method-A Jacobian linearization has error that grows quadratically in joint-speed×horizon (≈10 cm at 50% speed over 0.2 s), so fast or large-detour motions need a shorter horizon, an inflated Hessian-bound margin, or nonlinear rollout checking. **(iii) APF operating mode:** Method-B can settle at a local equilibrium; waypoint insertion or Method A is used when continued progress or a certified clearance is required. **(iv) Finite look-ahead:** the $N = 20$ (0.2 s) horizon covers the tested obstacle speeds; faster obstacles require a longer horizon or an earlier trigger. A terminal braking-to-rest safe set supports persistent feasibility. The evidence is simulation-based: randomized coverage comprises 20 FR3 obstacle trials, and the MuJoCo study is closed-loop rigid-body simulation rather than hardware.

B. Relationship to Path-First and Path–Velocity Pipelines

The path-first baseline (TOTG, TOPP-RA) is time-optimal by construction and plans the whole path at once, an advantage on static tasks. Its B2/B4 acceleration violations arise from circular-blend geometry, not time parameterization, and vanish with a C^2 path; the near-reversal singularity likewise eases once curvature is bounded. The architectural distinction is that a changed path triggers a batch rebuild and the pipeline begins from a precomputed geometric path, whereas the time-domain formulation updates the local trajectory directly. The two are complementary: the proposed planner trades global time-optimality and full-path lookahead for feasibility, smoothness, conditioning at degeneracies, and fixed-size reactive updates.

Industrial stacks such as Apollo separate path from speed planning along a fixed arc-length s, often with a virtual double-integrator speed model $\ddot{s} = u$ [11], and MoveIt pairs geometric planning with post-hoc time parameterization. The proposed method extends this virtual-integrator idea from timing along a fixed path to direct optimization of $q(t)$, updating path evolution, timing, kinematic limits, and local obstacle constraints in one convex QP.

A distinct line of reactive planners trades absolute time-optimality for high-frequency responsiveness: dynamical-system (DS) modulation [13] deflects a closed-form velocity field around obstacles, recent work pairs a joint-space DS with asynchronously triggered sampling-based MPC for non-convex obstacles [14], and MPPI [15] samples parallel rollouts. A parallel line keeps a precomputed path but governs the reference passed downstream: the Path Feasibility Governor [18] selects an auxiliary reference along the path that a nonlinear MPC is guaranteed to track with recursive feasibility and stability. The present planner modifies the reference in the same spirit but dispenses with the path altogether—there is no arc length to govern—trading that formal feasibility certificate for a fixed-structure convex QP and configuration-independent precomputation. The methods expose different guarantee sets: the present convex QP directly enforces hard joint/velocity/acceleration limits, while DS and sampling-based methods address broader non-convex avoidance. An asynchronous escape in the spirit of [14], triggered when the QP stalls, is a natural extension of the hybrid switch in Section IV-B. GPU batch optimizers operate at a different level: CuRobo [17] solves a *batch* global trajectory optimization in ≈ 50 ms on a GPU, complementary to a per-cycle receding-horizon planner—a natural deployment pairs a CuRobo-style global solver with the DI-QP for reactive correction. Compared with CHOMP, TrajOpt, nonlinear MPC, or whole-body MPC [12], which solve harder physical-dynamics problems, the present method instead treats the local planner as a virtual trajectory generator with precomputed Φ, Γ: a configuration-independent parameterization for high-rate replanning, not a new physical model.

IX. Conclusion

This paper presented a time-domain local Predictive Motion Planner on a virtual double-integrator backbone with constant A_d. Because the prediction matrices are precomputed, online planning reduces to a fixed-structure convex QP that directly enforces velocity, acceleration, and local obstacle constraints while producing C^1 reference trajectories. The experiments show the trade-off clearly: path-first TOTG/TOPP pipelines are faster on static paths, while the DI-QP keeps declared time-domain limits explicit and supports fixed-size reactive updates without a batch rebuild. Future work will focus on non-convex safe corridors, and comparisons with online MPC and trajectory optimization.

References

[1] I. A. Sucan, M. Moll, and L. E. Kavraki, "The Open Motion Planning Library," *IEEE Robotics & Automation Magazine*, vol. 19, no. 4, pp. 72–82, 2012.

[2] Q.-C. Pham and O. Stasse, "Time-optimal path parameterization for redundantly actuated robots: A numerical integration approach," *IEEE/ASME Transactions on Mechatronics*, vol. 20, no. 6, pp. 3257–3263, 2015.

[3] Y. Cao and J. Tang, "Toward Interaction Dynamics: A Predictive Framework for Safe Physical Human–Robot Interaction," *arXiv:2606.08281*, 2026. [Online]. Available: https://arxiv.org/abs/2606.08281

[4] B. Stellato, G. Banjac, P. Goulart, A. Bemporad, and S. Boyd, "OSQP: An operator splitting solver for quadratic programs," *Mathematical Programming Computation*, vol. 12, no. 4, pp. 637–672, 2020.

[5] S. Liu *et al.*, "Planning dynamically feasible trajectories for quadrotors using safe flight corridors in 3-D complex environments," *IEEE Robotics and Automation Letters*, vol. 2, no. 3, pp. 1688–1695, 2017.

[6] O. Khatib, "Real-time obstacle avoidance for manipulators and mobile robots," *International Journal of Robotics Research*, vol. 5, no. 1, pp. 90–98, 1986.

[7] H. Pham and Q.-C. Pham, "A new approach to time-optimal path parameterization based on reachability analysis," *IEEE Transactions on Robotics*, vol. 34, no. 3, pp. 645–659, 2018.

[8] O. Khatib, "A unified approach for motion and force control of robot manipulators: The operational space formulation," *IEEE Journal on Robotics and Automation*, vol. 3, no. 1, pp. 43–53, 1987.

[9] D. Q. Mayne *et al.*, "Constrained model predictive control: Stability and optimality," *Automatica*, vol. 36, no. 6, pp. 789–814, 2000.

[10] T. Kunz and M. Stilman, "Time-optimal trajectory generation for path following with bounded acceleration and velocity," in *Robotics: Science and Systems*, 2012.

[11] ApolloAuto, "Apollo: An open autonomous driving platform," GitHub repository, https://github.com/ApolloAuto/apollo, accessed June 17, 2026.

[12] J.-P. Sleiman, F. Farshidian, M. V. Minniti, and M. Hutter, "A unified MPC framework for whole-body dynamic locomotion and manipulation," *IEEE Robotics and Automation Letters*, vol. 6, no. 3, pp. 4688–4695, 2021.

[13] S. M. Khansari-Zadeh and A. Billard, "A dynamical system approach to realtime obstacle avoidance," *Autonomous Robots*, vol. 32, no. 4, pp. 433–454, 2012.

[14] M. Koptev, N. Figueroa, and A. Billard, "Reactive collision-free motion generation in joint space via dynamical systems and sampling-based MPC," *The International Journal of Robotics Research*, vol. 43, no. 13, pp. 2049–2069, 2024.

[15] G. Williams, N. Wagener, B. Goldfain, P. Drews, J. M. Rehg, B. Boots, and E. A. Theodorou, "Information-theoretic MPC for model-based reinforcement learning," in *IEEE Int. Conf. on Robotics and Automation (ICRA)*, 2017, pp. 1714–1721.

[16] M. Bhardwaj, B. Sundaralingam, A. Mousavian, N. D. Ratliff, D. Fox, F. Ramos, and B. Boots, "STORM: An integrated framework for fast joint-space model-predictive control for reactive manipulation," in *Proc. 5th Conf. on Robot Learning (CoRL)*, PMLR vol. 164, 2022, pp. 750–759.

[17] B. Sundaralingam, S. K. S. Hari, A. Fishman, C. Garrett, K. Van Wyk, V. Blukis, A. Millane, H. Oleynikova, A. Handa, F. Ramos, N. Ratliff, and D. Fox, "CuRobo: Parallelized collision-free robot motion generation," in *IEEE Int. Conf. on Robotics and Automation (ICRA)*, 2023, pp. 8112–8119.

[18] S. Zhang, J. Y. Z. Liu, and D. Liao-McPherson, "Integrating planning and predictive control using the path feasibility governor," *arXiv:2507.09134*, 2025. [Online]. Available: https://arxiv.org/abs/2507.09134